

## Chapter: Measures of Dispersion

Measures of dispersion or variation measures the extent of variation or deviation of individual values from central value. It helps us to measure variation among the data.

|                  |    |    |    |    |                |
|------------------|----|----|----|----|----------------|
| <b>Batsman 1</b> | 49 | 50 | 55 | 54 | $\bar{x} = 52$ |
| <b>Batsman 2</b> | 10 | 68 | 90 | 40 | $\bar{x} = 52$ |

Which batsman is more consistent?

### Objectives:

- To find variability among the data.
- To find the nature of distribution about the entire data.
- Find uniformness among the data.

### Types:

Different types of measures of dispersion or variation are:

1. The range
2. The interquartile range or quartile deviation
3. The mean deviation
4. The variance
5. The standard deviation

**Absolute Measures of Variation:** Absolute measures of variation are expressed in the same statistical unit in which the original data are given. **Ex:** Salary range between a manager and average salary of workers.

**Relative Measures of Variation:** A measure of relative variation is the ratio of a measure of absolute variation to an average. It also called coefficient of variation. **Ex:** The percentage of salary a manager get more from the average salary of workers.

## Absolute Measures of Variation

### Range:

The range is the absolute difference between the largest value and the smallest value in the set of data. Symbolically

$$\text{Range } R = L - S$$

Where  $L$  = Largest Value

$S$  = Smallest Value

**Example:** The following are the prices of shares of a company from Saturday to Thursday:

| Day       | Sat | Sun | Mon | Tue | Wed | Thu |
|-----------|-----|-----|-----|-----|-----|-----|
| Price(tk) | 200 | 210 | 208 | 160 | 220 | 250 |

**Solution:**

$$\text{Range } R = L - S = 250 - 160 = 90 \text{ tk}$$

✚ In the frequency distribution, range is calculated by taking the difference between the lower limit of the lowest class and upper limit of the highest class.

**Example:** Calculate the range of the following data

|                 |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 200   | 210   | 208   | 160   | 220   |

**Solution:**

$$\text{Range } R = L - S = 60 - 10 = 50 \text{ lakh}$$

### Limitation:

Range cannot tell us anything about the character in the distribution within two extreme observations.

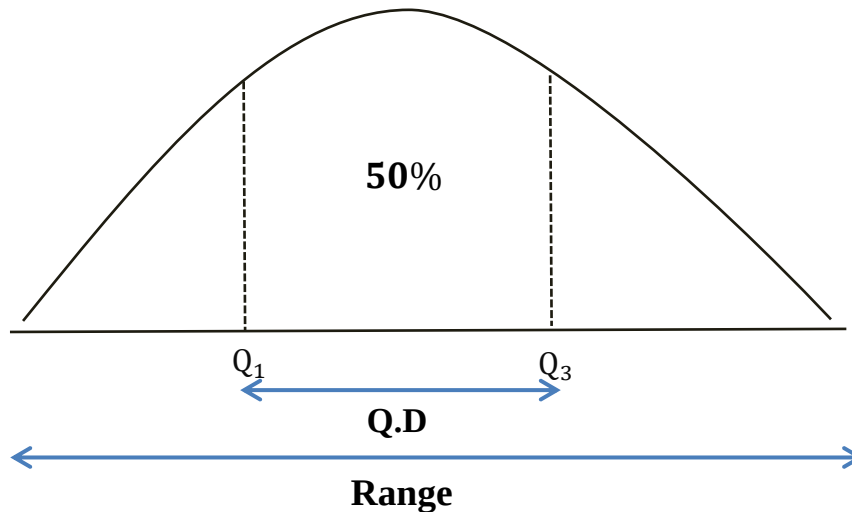
|               |   |    |    |    |    |    |    |             |
|---------------|---|----|----|----|----|----|----|-------------|
| <b>Data 1</b> | 6 | 46 | 46 | 46 | 46 | 46 | 46 | <b>R=40</b> |
| <b>Data 2</b> | 6 | 6  | 6  | 6  | 46 | 46 | 46 | <b>R=40</b> |
| <b>Data 3</b> | 6 | 10 | 15 | 25 | 36 | 39 | 46 | <b>R=40</b> |

✚ In all the three series of data range is same (i.e., 40), but it does not mean that the distribution of data are same.

## Interquartile range or quartile deviation:

Interquartile range or quartile deviation represents the difference between the third quartile  $Q_3$  and the first quartile  $Q_1$ . Symbolically

$$Q.D = \frac{Q_3 - Q_1}{2} \quad Q.D = \text{Quartile deviation}$$



✚ When quartile deviation is very small it describes small variation of the central 50% observations, and a high quartile deviation means that the variation among central observations is large.

$$\text{Quartiles } Q_i = L + \frac{\frac{i \times N}{4} - \text{p.c.f}}{f} \times h \quad i = 1, 2, 3$$

**Quartile class identified by  $Q_i = \frac{i \times N}{4}$  th observation.**

**Example:** The profits earned by 100 companies are given below:

| Profits (lakhs) | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 |
|-----------------|-------|-------|-------|-------|-------|-------|-------|--------|
| No of Companies | 4     | 8     | 18    | 30    | 15    | 10    | 8     | 7      |

- Calculate the range within which middle 50% companies fall.
- Calculate quartile deviation.

**Solution:**

| Profits (lakhs) | No of Companies | Cumulative frequency |
|-----------------|-----------------|----------------------|
| 20-30           | 4               | 4                    |
| 30-40           | 8               | 12                   |
| 40-50           | 18              | 30                   |
| 50-60           | 30              | 60                   |
| 60-70           | 15              | 75                   |
| 70-80           | 10              | 85                   |
| 80-90           | 8               | 93                   |
| 90-100          | 7               | 100                  |
| N = 100         |                 |                      |

The first quartile  $Q_1 = \frac{1 \times 100}{4} = 25$  th observation. 25 th observation lies in 40 - 50. Quartile class is 40 - 50.

We know  $Q_i = L + \frac{\frac{i \times N}{4} - \text{p.c.f}}{f} \times h$

$$Q_1 = 40 + \frac{\frac{1 \times 100}{4} - 12}{18} \times 10 = 47.22 \text{ lakhs}$$

The third quartile  $Q_3 = \frac{3 \times 100}{4} = 75$  th observation. 75 th observation lies in 60 - 70. Quartile class is 60 - 70.

$$Q_3 = 60 + \frac{\frac{3 \times 100}{4} - 60}{15} \times 10 = 70 \text{ lakhs}$$

Range within which middle 50% companies fall =  $Q_3 - Q_1 = 70 - 47.22 = 22.78$

Quartile deviation  $Q.D = \frac{Q_3 - Q_1}{2} = \frac{70 - 47.22}{2} = 11.39 \text{ lakhs}$

**Example:** Based on the frequency distribution given below calculate quartile deviation.

|                |      |       |       |       |       |       |       |
|----------------|------|-------|-------|-------|-------|-------|-------|
| Tax Paid(Lakh) | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
| No of Managers | 18   | 30    | 46    | 28    | 20    | 12    | 6     |

**ANS:**  $Q_1 = 13.67, Q_3 = 24.64, Q.D = 5.485$

**Example:** For the following data

|               |      |       |       |       |       |       |       |
|---------------|------|-------|-------|-------|-------|-------|-------|
| Age (yr)      | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
| No of Members | 6    | 5     | 8     | 15    | 7     | 6     | 3     |

- Calculate the range within which middle 50% members fall.
- Calculate quartile deviation.

### ***Limitation:***

Quartile deviation ignores 50% items, i.e., the first 25% and the last 25%. Since it does not depend upon every observation it is not regarded as good method of measuring variation.

### **Mean Deviation:**

Mean deviation is an average of absolute deviation of each observations from the mean. It is obtained by calculating the absolute deviation of each observations from the mean, and then averaging the deviations by taking their mean.

### ***Calculation of mean deviation (Ungrouped Data)***

For ungrouped data

$$M.D = \frac{\sum |x_i - \bar{x}|}{N}$$

**Example:** Based on the frequency distribution given below calculate mean deviation.

|                  |    |    |    |    |
|------------------|----|----|----|----|
| <b>Batsman 1</b> | 49 | 50 | 55 | 54 |
|------------------|----|----|----|----|

**Solution:**

**Batsman 1**

| $x_i$ | $\bar{x}$            | $ x_i - \bar{x} $           |
|-------|----------------------|-----------------------------|
| 49    | $\frac{208}{4} = 52$ | 3                           |
| 50    |                      | 2                           |
| 55    |                      | 3                           |
| 54    |                      | 2                           |
|       |                      | $\sum  x_i - \bar{x}  = 10$ |

→ |49-52|

$$M. D = \frac{\sum |x_i - \bar{x}|}{N} = \frac{10}{4} = 2.5$$

**Example:** Based on the frequency distribution given below calculate mean deviation.

|                  |    |    |    |    |
|------------------|----|----|----|----|
| <b>Batsman 2</b> | 10 | 68 | 90 | 40 |
|------------------|----|----|----|----|

**Solution:**

**Batsman 2**

| $x_i$ | $\bar{x}$            | $ x_i - \bar{x} $            |
|-------|----------------------|------------------------------|
| 10    | $\frac{208}{4} = 52$ | 42                           |
| 68    |                      | 16                           |
| 90    |                      | 38                           |
| 40    |                      | 12                           |
|       |                      | $\sum  x_i - \bar{x}  = 108$ |

$$M. D = \frac{\sum |x_i - \bar{x}|}{N} = \frac{108}{4} = 27$$

## Calculation of mean deviation (Grouped Data)

For grouped data

$$M. D = \frac{\sum f_i |x_i - \bar{x}|}{N}$$

Where

$\bar{x}$  = Arithmetic mean. ( $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h$ )

$x_i$  = Mid values of each class.

$N$  = The total frequency.

**Example:** Calculate mean deviation for the following data

|              |       |       |       |       |       |
|--------------|-------|-------|-------|-------|-------|
| Sales(Lakhs) | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of days   | 3     | 6     | 11    | 3     | 2     |

**Solution:**

| Sales | Mid value<br>$x_i$ | No of days<br>$f_i$ | $d_i$ | $f_i d_i$ | $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h$ | $ x_i - \bar{x} $ | $f_i  x_i - \bar{x} $            |
|-------|--------------------|---------------------|-------|-----------|-------------------------------------------------|-------------------|----------------------------------|
| 10-20 | 15                 | 3                   | -2    | -6        | $35 + \frac{-5}{25} \times 10$<br>$= 33$        | 18                | 54                               |
| 20-30 | 25                 | 6                   | -1    | -6        |                                                 | 8                 | 48                               |
| 30-40 | 35                 | 11                  | 0     | 0         |                                                 | 2                 | 22                               |
| 40-50 | 45                 | 3                   | +1    | +3        |                                                 | 12                | 36                               |
| 50-60 | 55                 | 2                   | +2    | +4        |                                                 | 22                | 44                               |
|       |                    | $N = 25$            |       |           |                                                 |                   | $\sum f_i  x_i - \bar{x}  = 204$ |

$$M. D = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{204}{25} = 8.16 \text{ lakhs}$$

**Example:** Calculate mean deviation for the following data

| Marks          | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
|----------------|------|-------|-------|-------|-------|-------|-------|
| No of Students | 6    | 5     | 8     | 15    | 7     | 6     | 3     |

**Solution:**

| Marks | Mid value<br>$x_i$ | $f_i$  | $d_i$               | $f_i d_i$ | $\bar{x}$                                  | $ x_i - \bar{x} $ | $f_i  x_i - \bar{x} $ |
|-------|--------------------|--------|---------------------|-----------|--------------------------------------------|-------------------|-----------------------|
| 0-10  | 5                  | 6      | -3                  | -18       | $35 + \frac{-8}{50} \times 10$<br>$= 33.4$ | 28.4              | 170.4                 |
| 10-20 | 15                 | 5      | -2                  | -10       |                                            | 18.4              | 92                    |
| 20-30 | 25                 | 8      | -1                  | -8        |                                            | 8.4               | 67.2                  |
| 30-40 | 35                 | 15     | 0                   | 0         |                                            | 1.6               | 24                    |
| 40-50 | 45                 | 7      | +1                  | +7        |                                            | 11.6              | 81.2                  |
| 50-60 | 55                 | 6      | +2                  | +12       |                                            | 21.6              | 129.6                 |
| 60-70 | 65                 | 3      | +3                  | +9        |                                            | 31.6              | 94.8                  |
|       |                    | N = 50 | $\sum f_i d_i = -8$ |           | $\sum f_i  x_i - \bar{x}  = 659.2$         |                   |                       |

$$M.D = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{659.2}{50} = 13.184$$

**Example:** Calculate mean deviation for the following data

| Class      | 0-6 | 6-12 | 12-18 | 18-24 | 24-30 |
|------------|-----|------|-------|-------|-------|
| No of days | 8   | 10   | 12    | 9     | 5     |

**ANS: 6.3**

**Limitation:**

In calculation algebraic signs are ignored while taking the deviations of the items. If the signs of the deviations are not ignored, the net sum of the deviations will be zero.



## Variance:

Variance is the average of the squares of the deviations of the given values from their arithmetic mean. It is denoted by  $\sigma^2$

### *Calculation of variance (Ungrouped Data)*

For ungrouped data, variance

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

**Example:** Find variance from the weekly wages of 10 workers working in a factory.

1320    1310    1315    1322    1326    1340    1325    1321    1320    1331

**Solution:**

| $x_i$ | $\bar{x}$                 | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$           |
|-------|---------------------------|-------------------|-------------------------------|
| 1320  | $\frac{13230}{10} = 1323$ | -3                | 9                             |
| 1310  |                           | -13               | 169                           |
| 1315  |                           | -8                | 64                            |
| 1322  |                           | -1                | 1                             |
| 1326  |                           | +3                | 9                             |
| 1340  |                           | +17               | 289                           |
| 1325  |                           | +2                | 4                             |
| 1321  |                           | -2                | 4                             |
| 1320  |                           | -3                | 9                             |
| 1331  |                           | +8                | 64                            |
|       |                           |                   | $\sum(x_i - \bar{x})^2 = 622$ |

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{622}{10} = 62.2 \text{ tk}$$

### *Calculation of variance (Grouped Data)*

For grouped data, variance

$$\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$$

or

$$\sigma^2 = \left[ \frac{\sum f_i d_i^2}{N} - \left( \frac{\sum f_i d_i}{N} \right)^2 \right] \times h^2$$

**Example:** Calculate variance for the following data

|                 |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8     | 12    | 20    | 6     | 4     |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$    | $d_i$ | $f_i d_i$ | $f_i d_i^2$                                |
|--------|--------------------|----------|-------|-----------|--------------------------------------------|
| 10-20  | 15                 | 8        | -2    | -16       | 32                                         |
| 20-30  | 25                 | 12       | -1    | -12       | 12                                         |
| 30-40  | 35                 | 20       | 0     | 0         | 0                                          |
| 40-50  | 45                 | 6        | +1    | +6        | 6                                          |
| 50-60  | 55                 | 4        | +2    | +8        | 16                                         |
|        |                    | $N = 50$ |       |           | $\sum f_i d_i = -14$ $\sum f_i d_i^2 = 66$ |

$f_1=8, d_1=-2$   
 $8 \times (-2)^2$

$$\sigma^2 = \left[ \frac{\sum f_i d_i^2}{N} - \left( \frac{\sum f_i d_i}{N} \right)^2 \right] \times h^2 = \left[ \frac{66}{50} - \left( \frac{-14}{50} \right)^2 \right] \times 10^2 =$$

$$= 1.2416 \times 100 = 124.16 \text{ lakhs}$$

**Example:** Calculate variance for the following data

|                 |      |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|
| Profit(Lakhs)   | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| No of Companies | 6    | 25    | 36    | 20    | 13    |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$     | $d_i$ | $f_i d_i$          | $f_i d_i^2$            |
|--------|--------------------|-----------|-------|--------------------|------------------------|
| 0-10   | 5                  | 6         | -2    | -12                | 24                     |
| 10-20  | 15                 | 25        | -1    | -25                | 25                     |
| 20-30  | 25                 | 36        | 0     | 0                  | 0                      |
| 30-40  | 35                 | 20        | +1    | +20                | 20                     |
| 40-50  | 45                 | 13        | +2    | +26                | 52                     |
|        |                    | $N = 100$ |       | $\sum f_i d_i = 9$ | $\sum f_i d_i^2 = 121$ |

$$\sigma^2 = \left[ \frac{\sum f_i d_i^2}{N} - \left( \frac{\sum f_i d_i}{N} \right)^2 \right] \times h^2 = \left[ \frac{121}{100} - \left( \frac{9}{100} \right)^2 \right] \times 10^2 =$$

$$= 1.2019 \times 100 = 120.19 \text{ lakhs}$$

**Example:** Calculate variance for the following data

|                 |      |       |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8    | 12    | 20    | 30    | 20    | 10    |

**ANS: 192.155**

**Example:** Calculate variance for the following data

|                   |      |       |       |       |       |       |       |
|-------------------|------|-------|-------|-------|-------|-------|-------|
| Tax<br>(Thousand) | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
| No of<br>Managers | 18   | 30    | 46    | 28    | 20    | 12    | 6     |

**ANS: 60.93**

## Standard Deviation:

Standard deviation is the positive square root of the average of the squares of the deviations of the given values from their arithmetic mean. It is denoted by  $\sigma$ . That is

$$\text{Standard deviation S.D or } \sigma = \sqrt{\sigma^2} = \sqrt{\text{variance}}$$

### *Calculation of standard deviation (Ungrouped Data)*

For ungrouped data, standard deviation

$$\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}}$$

**Example:** Find standard deviation from the weekly wages of 10 workers working in a factory.

1320   1310   1315   1322   1326   1340   1325   1321   1320   1331

**Solution:**

| $x_i$ | $\bar{x}$                 | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$             |
|-------|---------------------------|-------------------|---------------------------------|
| 1320  | $\frac{13230}{10} = 1323$ | -3                | 9                               |
| 1310  |                           | -13               | 169                             |
| 1315  |                           | -8                | 64                              |
| 1322  |                           | -1                | 1                               |
| 1326  |                           | +3                | 9                               |
| 1340  |                           | +17               | 289                             |
| 1325  |                           | +2                | 4                               |
| 1321  |                           | -2                | 4                               |
| 1320  |                           | -3                | 9                               |
| 1331  |                           | +8                | 64                              |
|       |                           |                   | $\Sigma(x_i - \bar{x})^2 = 622$ |

$$\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}} = \sqrt{\frac{622}{10}} = \sqrt{62.2} = 7.89$$

## Calculation of standard deviation (Grouped Data)

For grouped data, standard deviation

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$$

or

$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h$$

**Example:** Calculate standard deviation for the following data

|                 |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8     | 12    | 20    | 6     | 4     |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$    | $d_i$ | $f_i d_i$ | $f_i d_i^2$          |                                      |
|--------|--------------------|----------|-------|-----------|----------------------|--------------------------------------|
| 10-20  | 15                 | 8        | -2    | -16       | 32                   | $f_1=8, d_1=-2$<br>$8 \times (-2)^2$ |
| 20-30  | 25                 | 12       | -1    | -12       | 12                   |                                      |
| 30-40  | 35                 | 20       | 0     | 0         | 0                    |                                      |
| 40-50  | 45                 | 6        | +1    | +6        | 6                    |                                      |
| 50-60  | 55                 | 4        | +2    | +8        | 16                   |                                      |
|        |                    | $N = 50$ |       |           | $\sum f_i d_i = -14$ | $\sum f_i d_i^2 = 66$                |

$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{66}{50} - \left(\frac{-14}{50}\right)^2} \times 10$$

$$= 11.14 \text{ lakhs}$$

**Example:** Calculate standard deviation for the following data

|                 |      |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|
| Profit(Lakhs)   | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| No of Companies | 6    | 25    | 36    | 20    | 13    |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$     | $d_i$ | $f_i d_i$          | $f_i d_i^2$            |
|--------|--------------------|-----------|-------|--------------------|------------------------|
| 0-10   | 5                  | 6         | -2    | -12                | 24                     |
| 10-20  | 15                 | 25        | -1    | -25                | 25                     |
| 20-30  | 25                 | 36        | 0     | 0                  | 0                      |
| 30-40  | 35                 | 20        | +1    | +20                | 20                     |
| 40-50  | 45                 | 13        | +2    | +25                | 52                     |
|        |                    | $N = 100$ |       | $\sum f_i d_i = 9$ | $\sum f_i d_i^2 = 121$ |

$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{121}{100} - \left(\frac{9}{100}\right)^2} \times 10$$

$$= 10.96 \text{ lakhs}$$

**Example:** An analysis of production rejects resulted in the following data

|                 |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|
| Reject amount   | 21-25 | 26-30 | 31-35 | 36-40 | 41-45 | 46-50 | 51-55 |
| No of Operators | 5     | 15    | 28    | 42    | 15    | 12    | 3     |

Calculate mean and standard deviation.

**Solution:** Using class boundaries we get

| Reject amount | Mid value<br>$x_i$ | $f_i$     | $d_i$ | $f_i d_i$            | $f_i d_i^2$            |
|---------------|--------------------|-----------|-------|----------------------|------------------------|
| 20.5-25.5     | 23                 | 5         | -3    | -15                  | 45                     |
| 25.5-30.5     | 28                 | 15        | -2    | -30                  | 60                     |
| 30.5-35.5     | 33                 | 28        | -1    | -28                  | 28                     |
| 35.5-40.5     | 38                 | 42        | 0     | 0                    | 0                      |
| 40.5-45.5     | 43                 | 15        | +1    | +15                  | 15                     |
| 45.5-50.5     | 48                 | 12        | +2    | +24                  | 48                     |
| 50.5-55.5     | 53                 | 3         | +3    | +9                   | 27                     |
|               |                    | $N = 120$ |       | $\sum f_i d_i = -25$ | $\sum f_i d_i^2 = 223$ |

Mean:  $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h = 38 + \frac{-25}{120} \times 10 = 36.96$

Standard deviation:  $\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{223}{120} - \left(\frac{-25}{120}\right)^2} \times 5$   
 $= \sqrt{1.858 - .043} \times 5 = 6.375$

**Example:** Calculate standard deviation for the following data

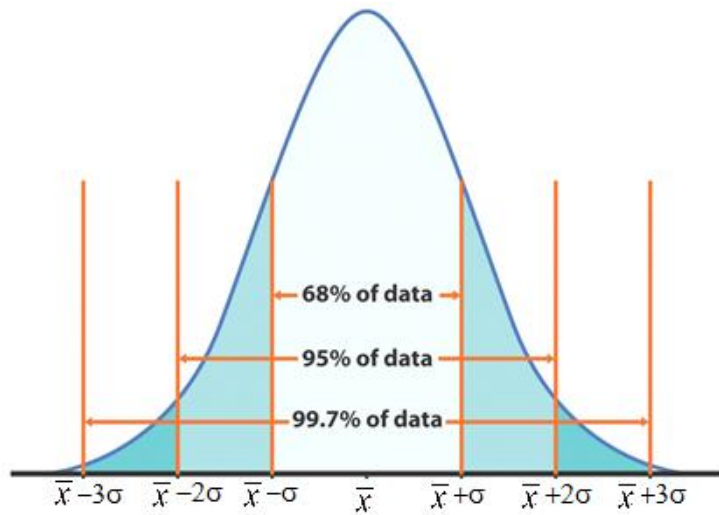
|                 |      |       |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8    | 12    | 20    | 30    | 20    | 10    |

**ANS: 13.862**

**Example:** Calculate standard deviation for the following data

|                   |      |       |       |       |       |       |       |
|-------------------|------|-------|-------|-------|-------|-------|-------|
| Tax<br>(Thousand) | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
| No of<br>Managers | 18   | 30    | 46    | 28    | 20    | 12    | 6     |

**ANS: 7.806**



- $\bar{x} \pm \sigma$  covers 68.27% observations.
- $\bar{x} \pm 2\sigma$  covers 95.45% observations.
- $\bar{x} \pm 3\sigma$  covers 99.73% observations.

**Example:** The breaking strength of 80 test pieces of a certain alloy is give in the following table, the unit being given to the nearest pounds per square inch.

|                   |       |       |       |       |       |
|-------------------|-------|-------|-------|-------|-------|
| Breaking strength | 44-46 | 46-48 | 48-50 | 50-52 | 52-54 |
| No of pieces      | 3     | 24    | 27    | 21    | 5     |

Calculate the average breaking strength of the alloy and the standard deviation. Calculate the percentage of observations lying between mean  $\pm 2\sigma$ .

**Solution:**

| Breaking strength | Mid value $x_i$ | $f_i$    | $d_i$ | $f_i d_i$          | $f_i d_i^2$           |
|-------------------|-----------------|----------|-------|--------------------|-----------------------|
| 44-46             | 45              | 3        | -2    | -6                 | 12                    |
| 46-48             | 47              | 24       | -1    | -24                | 24                    |
| 48-50             | 49              | 27       | 0     | 0                  | 0                     |
| 50-52             | 51              | 21       | +1    | +21                | 21                    |
| 52-54             | 53              | 5        | +2    | +10                | 20                    |
|                   |                 | $N = 80$ |       | $\sum f_i d_i = 1$ | $\sum f_i d_i^2 = 77$ |

$$\bar{x} = A + \frac{\sum f_i d_i}{N} \times h = 49 + \frac{1}{80} \times 2 = 49.025$$



$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{77}{80} - \left(\frac{1}{80}\right)^2} \times 2$$

$$= 1.96$$

Now  $\bar{x} \pm 2\sigma = 49.025 \pm 2(1.96) = 45.105$  to  $52.945 = 45$  to  $53$  (approx.). We have to calculate the percentage of items lying between 45 to 53. Since the number are equally distributed, then the frequency at 45 be 1.5 and the frequency at 53 be 2.5.

Thus the total frequency between 45 to 53 is  $(1.5 + 24 + 27 + 21 + 2.5) = 76$ . The percentage is  $\frac{76}{80} \times 100 = 95$ . Thus there are 95 percent observations lying between mean  $\pm 2\sigma$ .

## Relative Measures of Variation

### Coefficient of Variance:

Measures of variance discussed above are absolute measures. One of the relative measure is known as coefficient of variance. A coefficient of variance is computed as a percentage of the standard deviation of the distribution to the mean of the same distribution. This measure represents the **spread of the distribution relative to the mean** of the same distribution. Symbolically:

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

Where  $\sigma$  = Standard deviation

$\bar{x}$  = Arithmetic mean

Consider a statistic of height and weight of the 150 children:

|             | Height       | Weight     |
|-------------|--------------|------------|
| <b>Mean</b> | 40 inch      | 10 kg      |
| <b>S.D</b>  | 5 inch       | 2 kg       |
| <b>C.V</b>  | <b>12.5%</b> | <b>20%</b> |

**Which one has more variability among the children, height or weight?**

✚ The coefficient of variance helpful in comparing the relative variation in several data set having different measure of unit.

Consider the blood pressure of a group of patient were measured at two levels:

|                | Systolic (mm) | Diastolic (mm) |
|----------------|---------------|----------------|
| <b>Average</b> | 130           | 60             |
| <b>S.D</b>     | 15            | 8              |
| <b>C.V</b>     | <b>11.5%</b>  | <b>13.3%</b>   |

Which pressure shows greater variability among the patients?

✚ The coefficient of variance helpful in comparing the relative variation in several data set having different means and different standard deviations.

### *Calculation of coefficient of variance (Ungrouped Data)*

For ungrouped data, coefficient of variance

$$C. V = \frac{\sigma}{\bar{X}} \times 100$$

Where  $\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}}$  and  $\bar{x} = \frac{\sum x_i}{N}$

**Example:** Find coefficient of variance from the price of a company share during the last 10 months in Dhaka stock exchange.

105   120   115   118   130   127   109   110   104   112

**Solution:**

| $x_i$ | $\bar{x}$               | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$             |
|-------|-------------------------|-------------------|---------------------------------|
| 105   | $\frac{1150}{10} = 115$ | -10               | 100                             |
| 120   |                         | +5                | 25                              |
| 115   |                         | 0                 | 0                               |
| 118   |                         | +3                | 9                               |
| 130   |                         | +15               | 225                             |
| 127   |                         | +12               | 144                             |
| 109   |                         | -6                | 36                              |
| 110   |                         | -5                | 25                              |
| 104   |                         | -11               | 121                             |
| 112   |                         | -3                | 9                               |
|       |                         |                   | $\Sigma(x_i - \bar{x})^2 = 694$ |

Standard deviation  $\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}} = \sqrt{\frac{694}{10}} = \sqrt{69.4} = 8.33 \text{ tk}$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{8.33}{115} \times 100 = 7.24 \%$$

**Example:** Find coefficient of variance from the price of a company share during the last 10 months in Chittagong stock exchange.

108 117 120 130 100 125 125 120 110 135

**Solution:**

| $x_i$ | $\bar{x}$               | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$              |
|-------|-------------------------|-------------------|----------------------------------|
| 108   | $\frac{1190}{10} = 119$ | -11               | 121                              |
| 117   |                         | -2                | 4                                |
| 120   |                         | +1                | 1                                |
| 130   |                         | +11               | 121                              |
| 100   |                         | -19               | 361                              |
| 125   |                         | +6                | 36                               |
| 125   |                         | +6                | 36                               |
| 120   |                         | +1                | 1                                |
| 110   |                         | -9                | 81                               |
| 135   |                         | +16               | 256                              |
|       |                         |                   | $\Sigma(x_i - \bar{x})^2 = 1018$ |

Standard deviation  $\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}} = \sqrt{\frac{1018}{10}} = \sqrt{101.8} = 10.09 \text{ tk}$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{10.09}{119} \times 100 = 10.09 \%$$

➤ Since the coefficient of variance of Dhaka is less, therefore it is safe to invest in Dhaka stock exchange.

### Calculation of coefficient of variance (Grouped Data)

For grouped data, coefficient of variance

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

Where  $\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h$  and  $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h$

**Example:** Calculate standard deviation and coefficient of variance for the following data

|                 |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8     | 12    | 20    | 6     | 4     |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$    | $d_i$ | $f_i d_i$            | $f_i d_i^2$           |
|--------|--------------------|----------|-------|----------------------|-----------------------|
| 10-20  | 15                 | 8        | -2    | -16                  | 32                    |
| 20-30  | 25                 | 12       | -1    | -12                  | 12                    |
| 30-40  | 35                 | 20       | 0     | 0                    | 0                     |
| 40-50  | 45                 | 6        | +1    | +6                   | 6                     |
| 50-60  | 55                 | 4        | +2    | +8                   | 16                    |
|        |                    | $N = 50$ |       | $\sum f_i d_i = -14$ | $\sum f_i d_i^2 = 66$ |

Mean  $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h = 35 + \frac{-14}{50} \times 10 = 32.2$  lakh

Standard deviation  $\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{66}{50} - \left(\frac{-14}{50}\right)^2} \times 10 = 11.14$  lakh

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{11.14}{32.2} \times 100 = 34.6 \%$$

**Example:** Calculate standard deviation and coefficient of variance for the following data

|                 |      |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|
| Profit(Lakhs)   | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| No of Companies | 6    | 25    | 36    | 20    | 13    |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$     | $d_i$ | $f_i d_i$          | $f_i d_i^2$            |
|--------|--------------------|-----------|-------|--------------------|------------------------|
| 0-10   | 5                  | 6         | -2    | -12                | 24                     |
| 10-20  | 15                 | 25        | -1    | -25                | 25                     |
| 20-30  | 25                 | 36        | 0     | 0                  | 0                      |
| 30-40  | 35                 | 20        | +1    | +20                | 20                     |
| 40-50  | 45                 | 13        | +2    | +25                | 52                     |
|        |                    | $N = 100$ |       | $\sum f_i d_i = 9$ | $\sum f_i d_i^2 = 121$ |

Mean:  $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h = 25 + \frac{9}{100} \times 10 = 25.9 \text{ lakh}$

Standard deviation:  $\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{121}{100} - \left(\frac{9}{100}\right)^2} \times 10 = 10.96 \text{ lakh}$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{10.96}{25.9} \times 100 = 42.32 \%$$

**Example:** An analysis of production rejects resulted in the following data

|                 |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|
| Reject amount   | 21-25 | 26-30 | 31-35 | 36-40 | 41-45 | 46-50 | 51-55 |
| No of Operators | 5     | 15    | 28    | 42    | 15    | 12    | 3     |

Calculate standard deviation and coefficient of variance.

**Solution:** Using class boundaries we get

| Reject amount | Mid value<br>$x_i$ | $f_i$     | $d_i$ | $f_i d_i$            | $f_i d_i^2$            |
|---------------|--------------------|-----------|-------|----------------------|------------------------|
| 20.5-25.5     | 23                 | 5         | -3    | -15                  | 45                     |
| 25.5-30.5     | 28                 | 15        | -2    | -30                  | 60                     |
| 30.5-35.5     | 33                 | 28        | -1    | -28                  | 28                     |
| 35.5-40.5     | 38                 | 42        | 0     | 0                    | 0                      |
| 40.5-45.5     | 43                 | 15        | +1    | +15                  | 15                     |
| 45.5-50.5     | 48                 | 12        | +2    | +24                  | 48                     |
| 50.5-55.5     | 53                 | 3         | +3    | +9                   | 27                     |
|               |                    | $N = 120$ |       | $\sum f_i d_i = -25$ | $\sum f_i d_i^2 = 223$ |

Mean:  $\bar{x} = A + \frac{\sum f_i d_i}{N} \times h = 38 + \frac{-25}{120} \times 10 = 36.96$

Standard deviation: 
$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{223}{120} - \left(\frac{-25}{120}\right)^2} \times 5$$
$$= \sqrt{1.858 - .043} \times 5 = 6.375$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{6.375}{36.96} \times 100 = 17.25 \%$$

## **For Practice**

1. Calculate standard deviation and coefficient of variance for the following data

|                 |      |       |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|-------|
| Profit          | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8    | 12    | 20    | 30    | 20    | 10    |

**ANS:  $\bar{x} = 32.2, \sigma = 13.862, C.V = 43.05\%$**

2. Calculate standard deviation and coefficient of variance for the following data

|                |      |       |       |       |       |       |       |
|----------------|------|-------|-------|-------|-------|-------|-------|
| Tax (Thou)     | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
| No of Managers | 18   | 30    | 46    | 28    | 20    | 12    | 6     |

**ANS:  $\bar{x} = 17.6, \sigma = 7.806, C.V = 44.35\%$**

3. Calculate standard deviation and coefficient of variance for the following data

|               |       |       |       |       |       |       |       |       |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Wages         | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 |
| No of Workers | 1     | 2     | 4     | 13    | 21    | 9     | 6     | 4     |

**ANS:  $\bar{x} = 55.33, \sigma = 14.83, C.V = 26.8\%$**

4. Calculate standard deviation and coefficient of variance for the following data

|                 |       |       |       |       |       |       |       |        |
|-----------------|-------|-------|-------|-------|-------|-------|-------|--------|
| Profits         | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 |
| No of Companies | 4     | 8     | 18    | 30    | 15    | 10    | 8     | 7      |

**ANS:  $\bar{x} = 59.1, \sigma = 17.56, C.V = 29.71\%$**

5. Calculate standard deviation and coefficient of variance for the following data

|            |         |         |         |         |         |         |         |         |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|
| Weights    | 210-215 | 215-220 | 220-225 | 225-230 | 230-235 | 235-240 | 240-245 | 245-250 |
| No of Boys | 8       | 13      | 16      | 29      | 14      | 10      | 7       | 3       |

**ANS:  $\bar{x} = 227.55, \sigma = 8.732, C.V = 3.84\%$**

6. Calculate standard deviation and coefficient of variance for the following data

|                 |      |       |       |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|-------|-------|
| Turnover(Lakh)  | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
| No of Companies | 8    | 18    | 42    | 62    | 30    | 10    | 4     |

**ANS:  $\bar{x} = 21.35, \sigma = 6.375, C.V = 29.86\%$**

## Empirical Relation between Measures of variation

$$\text{Quartile Deviation (Q.D)} = \frac{2}{3} \text{ Standard Deviation } (\sigma)$$

$$\text{Mean Deviation (M.D)} = \frac{4}{5} \text{ Standard Deviation } (\sigma)$$

$$\text{Quartile Deviation (Q.D)} = \frac{5}{6} \text{ Mean Deviation (M.D)}$$

**Example:** Calculate standard deviation and then calculate mean deviation using empirical relation for the following data

|                 |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|
| Profit(Lakhs)   | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
| No of Companies | 8     | 12    | 20    | 6     | 4     |

**Solution:**

| Profit | Mid value<br>$x_i$ | $f_i$    | $d_i$ | $f_i d_i$ | $f_i d_i^2$           |
|--------|--------------------|----------|-------|-----------|-----------------------|
| 10-20  | 15                 | 8        | -2    | -16       | 32                    |
| 20-30  | 25                 | 12       | -1    | -12       | 12                    |
| 30-40  | 35                 | 20       | 0     | 0         | 0                     |
| 40-50  | 45                 | 6        | +1    | +6        | 6                     |
| 50-60  | 55                 | 4        | +2    | +8        | 16                    |
|        |                    | $N = 50$ |       |           | $\sum f_i d_i = -14$  |
|        |                    |          |       |           | $\sum f_i d_i^2 = 66$ |

Standard deviation:  $\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times h = \sqrt{\frac{66}{50} - \left(\frac{-14}{50}\right)^2} \times 10 = 11.14 \text{ lakh}$

Mean deviation:  $M.D = \frac{4}{5} \times \sigma = \frac{4}{5} \times 11.14 = 8.912 \text{ lakh}$



**Example:** Calculate mean deviation and then calculate quartile deviation using empirical relation for the following data

| Marks          | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
|----------------|------|-------|-------|-------|-------|-------|-------|
| No of Students | 6    | 5     | 8     | 15    | 7     | 6     | 3     |

**Solution:**

| Marks | Mid value<br>$x_i$ | $f_i$  | $d_i$ | $f_i d_i$           | $\bar{x}$                                  | $ x_i - \bar{x} $ | $f_i  x_i - \bar{x} $              |
|-------|--------------------|--------|-------|---------------------|--------------------------------------------|-------------------|------------------------------------|
| 0-10  | 5                  | 6      | -3    | -18                 | $35 + \frac{-8}{50} \times 10$<br>$= 33.4$ | 28.4              | 170.4                              |
| 10-20 | 15                 | 5      | -2    | -10                 |                                            | 18.4              | 92                                 |
| 20-30 | 25                 | 8      | -1    | -8                  |                                            | 8.4               | 67.2                               |
| 30-40 | 35                 | 15     | 0     | 0                   |                                            | 1.6               | 24                                 |
| 40-50 | 45                 | 7      | +1    | +7                  |                                            | 11.6              | 81.2                               |
| 50-60 | 55                 | 6      | +2    | +12                 |                                            | 21.6              | 129.6                              |
| 60-70 | 65                 | 3      | +3    | +9                  |                                            | 31.6              | 94.6                               |
|       |                    | N = 50 |       |                     |                                            |                   | $\sum f_i  x_i - \bar{x}  = 658.4$ |
|       |                    |        |       | $\sum f_i d_i = -8$ |                                            |                   |                                    |

$$M.D = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{658.4}{50} = 13.168$$

Quartile deviation       $Q.D = \frac{5}{6} \times M.D = \frac{5}{6} \times 13.168 = 10.97$